# Yuliang(Leo) Chen

@ yuliang.chen.gr@dartmouth.edu | GitHub | Portfolio

## Education

**Dartmouth College, Hanover** — Hanover, New Hampshire
*Ph.D. Student in Computer Science* — *Sep 2025 – Jun 2030*

**University of California, San Diego** — San Diego, California
*M.S. in Data Science* — *Sep 2023 – Jun 2025*

**University of California, Santa Barbara** — Santa Barbara, California
*B.S. in Statistics – Data Science | B.S. in Mathematics* — *Sep 2018 – Jun 2022*

**Research Interests:** LLM post-training and reasoning, multimodal representation learning, self-supervised learning, LLM agents/agentic systems, AI4Health (Bio Sensing)

## Publications

**Lapras: Latent Reasoning for Time Series Language Models**
**Yuliang Chen**, Yu Yvonne Wu, Patrick Langer, Arvind Pillai, Sudarshan Regmi, Martin Maritsch, Juncheng Liu, Robert Jakob, Thomas Kaar, Tess Z. Griffin, Lisa Marsch, Michael V. Heinz, Nicholas C. Jacobson, and Andrew Campbell. *Under review*, 2026.

**Learning Transferable Sensor Models via Language-Informed Pretraining**
**Yuliang Chen**, Arvind Pillai, Yu Yvonne Wu, Tess Z. Griffin, Lisa Marsch, Michael V. Heinz, Nicholas C. Jacobson, and Andrew Campbell. *arXiv preprint arXiv:2603.11950*, 2026.

**Toward Foundation Model for Multivariate Wearable Sensing of Physiological Signals**
Yunfei Luo*, **Yuliang Chen***, Asif Salekin, and Tauhidur Rahman. *ACM Transactions on Computing for Healthcare (HEALTH)*, 2025.

**Multi-modal Cross-masked Autoencoder for Digital Health Measurements**
Howon Ryu*, **Yuliang Chen***, Yacun Wang, Andrea Z. LaCroix, Loki Natarajan, Yu Wang, and Jingjing Zou. *arXiv preprint arXiv:2506.02260*, 2025.

**BALMS: Benchmarking Agentic LLMs for Longitudinal Mental Health Sensing**
Yu Yvonne Wu, Arvind Pillai, **Yuliang Chen**, Yuwei Zhang, Sudarshan Regmi, Tess Z. Griffin, Michael V. Heinz, Lisa A. Marsch, Nicholas C. Jacobson, and Andrew Campbell. *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.

**Toward World Modeling of Physiological Signals with Chaos-Theoretic Balancing and Latent Dynamics**
Yunfei Luo, Xi Chen, **Yuliang Chen**, Lanshuang Zhang, Md Mofijul Islam, Siwei Zhao, Peter Kotanko, Subhasis Dasgupta, Andrew Campbell, Rakesh Malhotra, and Tauhidur Rahman. *arXiv preprint arXiv:2605.15465*, 2026.

* *Equal contribution; order decided by a coin flip.*

## Research Experience

**HealthX Lab, Dartmouth College** — Sep 2025 – Present
*Graduate Research Assistant; Advisor: Prof. Andrew Campbell* — *Hanover, NH*

- Developed **Lapras**, a post-training framework that enables LLMs (e.g., Qwen3-8B) to reason over time series in latent representation space rather than explicit text, improving average F1 by **10.8%** over standard chain-of-thought fine-tuning while generating **23.9× fewer tokens**.
- Pretrained **SLIP**, a **unified sensor–language model** that learns a shared sensor–text embedding space from **heterogeneous** sensor data (wearable, IoT, energy, environmental) on **600K sensor–caption pairs**, surpassing the state-of-the-art time series foundation model by **5.7%** in linear-probe accuracy across **11 datasets**.
- **Instruction-tuned** SLIP for open-ended sensor question answering and captioning, exceeding the state-of-the-art time series language model by **7.5%** in average QA accuracy across 4 benchmarks.
- Co-developed **BALMS**, the first **benchmark of LLM agents** for wellbeing prediction from longitudinal wearable data, comparing prompt-based, tool-augmented, and RAG agents across 5 LLMs and 3 real-world datasets.

**Zou Lab, UC San Diego** Jul 2024 – Sep 2025

*Graduate Student Researcher; Advisor: Prof. Jingjing Zou* *San Diego, CA*

- Pretrained **MoCA**, a theoretically grounded **multimodal masked autoencoder** for longitudinal wearable data, improving linear-probe accuracy by **15.0%** over prior methods across 6 transfer benchmarks.
- Shipped an open-source **lightweight sitting-detection model** for wrist sensors, building an end-to-end pipeline over **8.2K hours** of real-world data and improving balanced accuracy by **10.7%** over the prior published model.

**MoSAIC Lab, UC San Diego** Mar 2024 – Jun 2025

*Research Assistant; Advisor: Prof. Tauhidur Rahman* *San Diego, CA*

- Pretrained **NormWear**, a **sensor-agnostic foundation model** for multivariate wearable signals, outperforming the state-of-the-art time series foundation model by **6.1%** in AUC-ROC across **18 tasks and 11 datasets**.
- Co-developed **NormWear-2**, a **world model** that captures physiological signals and clinical interventions through shared latent dynamics, achieving **up to 13.5%** higher composite forecasting score than state-of-the-art time series foundation models across 5 datasets spanning millisecond to hourly resolutions.

## Skills

**Languages:** Python, SQL, R, C/C++ **Infra:** Slurm, Ray, Kubernetes, Docker, Hydra, Weights & Biases

**Training:** PyTorch (DDP, FSDP), DeepSpeed (ZeRO), HF Accelerate, TRL, verl, LLaMA-Factory, vLLM